AXI4-Lite Firewall IP Core

User Guide

IP core	`altera_axi4_lite_firewall`
Core version	2.0
Document version	1.0
Date	August 2026
Target	Intel FPGA / Quartus Prime, Platform Designer

Contents

1. About the AXI4-Lite Firewall IP Core

The AXI4-Lite Firewall IP core is a bump-in-the-wire component that sits between a bus master and a peripheral you want to protect. It enforces an allow-list on every transaction and guarantees the master always receives a response, even when the protected peripheral stops answering.

There is no equivalent core in the Intel FPGA IP catalog. AMD/Xilinx ship an AXI Protocol Firewall, but that core polices protocol legality and has no address-range access control; the two are complementary rather than equivalent. See Section 9 for a detailed comparison.

The core is delivered as synthesisable SystemVerilog with a Platform Designer component description, a self-checking testbench, SystemVerilog assertions, and both a Questa and a licence-free Verilator simulation flow.

1.1 Features

- **Address-range access control.** Up to 64 software-programmable address ranges, each with independent read and write permission. Default-deny: a transaction matching no rule is rejected.
- **Fault isolation.** A programmable round-trip timeout on every forwarded transaction. On expiry the core answers the master itself, so a wedged peripheral cannot hang the bus master.
- **Separate control port.** Configuration and status live on a physically distinct AXI4-Lite slave, so recovery is never blocked by the fault being recovered from.
- **Interrupt output.** Level interrupt, asserted while any enabled sticky fault bit is set.
- **Isolation control.** Software can isolate the peripheral manually, and the core can isolate it automatically on a timeout.
- **Full status visibility.** Outstanding-transaction and stuck-command status bits let a driver sequence recovery correctly.
- **Secure by default.** Access control is enabled and the rule table is empty out of reset, so an unconfigured core denies everything rather than passing everything.

1.2 Device Family Support

The core is written in device-independent synthesisable SystemVerilog. It instantiates no family-specific primitives, no vendor macros, no PLLs and no memory blocks — the rule table is implemented in registers. It should therefore compile for any Intel FPGA family supported by your Quartus Prime installation.

Table 1. Device Family Support

Device family	Support level
All Intel FPGA families	Expected to compile; **not characterised**

> **Note**
> No device family has been characterised. The core has not been compiled by Quartus Prime at the time of writing — it has been verified only by simulation and static elaboration. See Section 8 for exactly what has and has not been established.

1.3 Resource Utilization

Not characterised. No synthesis has been run, so no logic element, register or memory figures are available.

The two structures that dominate area and timing are predictable from the source, and are recorded here so you know what to expect and what to measure:

- **The rule table** is `NUM_RULES × (2 × ADDR_WIDTH + 3)` registers. At the default `NUM_RULES = 8` and `ADDR_WIDTH = 32` that is 536 registers.
- **The rule lookup** is a purely combinational priority chain over `NUM_RULES` entries, duplicated for the read and write paths. Each entry is two `ADDR_WIDTH`-wide magnitude comparisons. This is the expected critical path and the term that scales with `NUM_RULES`.

To obtain real figures, synthesise the core standalone with `quartus_map`, then run `quartus_sta` for f_{MAX}, sweeping `NUM_RULES`. If the lookup limits f_{MAX}, the standard remedy is to register the lookup result and accept one extra cycle of latency in the EVAL state.

1.4 Release Information

Table 2. Release Information

Item	Description
Version	2.0
`CORE_INFO` version field	`0x0200`
Release date	August 2026
Ordering code	None — source-available
Vendor	monkstein88
IP catalog name	AXI4-Lite Firewall
IP catalog group	Bridges and Adapters / Custom
Language	SystemVerilog (IEEE 1800)

> **Caution**
> Version 2.0 is a breaking change from 1.x. The peripheral reset output was removed and the recovery procedure changed. A 1.x driver run against a 2.0 core will acknowledge a fault and then see every subsequent transaction return SLVERR. Read the version field in `CORE_INFO` before assuming either behaviour. See Section 7.5.

2. Getting Started

2.1 Installing the IP Core

The core is distributed as source. To make it visible to Platform Designer:

1. Clone or copy the repository to your machine.
2. In Quartus Prime, click **Tools ▸ Options ▸ IP Catalog Search Locations**.
3. Add the repository's **top-level directory** — the one containing `altera_axi4_lite_firewall/` — to the search path.
4. In Platform Designer, click **File ▸ Refresh System**.

The core appears in the IP Catalog as **AXI4-Lite Firewall** under *Bridges and Adapters / Custom*.

> **Note**
> `_hw.tcl` syntax has changed across Quartus Prime releases, and between Standard and Pro editions. The supplied `axi_firewall_hw.tcl` has not been validated against any specific release. If the component fails to import, see Section 2.1.1.

2.1.1 If the Component Does Not Import

Package the component manually. This takes a few minutes and produces a `_hw.tcl` guaranteed correct for your installed toolchain:

1. In Platform Designer, click **File ▸ New Component**.
2. On the **Files** tab, add `rtl/axi_firewall_regs.sv` and `rtl/axi_firewall_top.sv` as synthesis files. Set `axi_firewall_top.sv` as the top-level file.
3. Click **Analyze Synthesis Files**.
4. On the **Signals & Interfaces** tab, confirm the grouping. Every port follows the `s_axi_*` / `m_axi_*` / `s_axi_ctrl_*` convention with standard AXI4-Lite suffixes, so signal analysis should infer three AXI4-Lite interfaces plus clock, reset and interrupt automatically. Correct anything it misses.
5. Click **Finish** and save.

> **Caution**
> Declare the RTL as **SystemVerilog**, not Verilog. The source uses `logic`, `always_ff`, packed structs and enumerated types. Analysed with the Verilog-2001 parser it fails on the first `logic` declaration.

2.2 Adding the Core to a Platform Designer System

Instantiate the core between the master and the peripheral to be protected:

- Connect the master's data master to `s_axi` and to `s_axi_ctrl`.
- Connect `m_axi` to the protected peripheral.
- Connect `irq` to an interrupt input on the CPU.
- Connect `clock` and `reset` to the system clock and reset.

No explicit bridge is required between Avalon-MM and AXI4-Lite. Platform Designer's interconnect provides the bridging logic. An *AXI Bridge Intel FPGA IP* may be instantiated explicitly if you want to trade concurrency for reduced interconnect logic, but it is optional.

Assign `s_axi_ctrl` a base address in the CPU's address map. The rule addresses programmed into the core refer to addresses as seen on `s_axi`, which is the protected peripheral's address decode.

2.3 Required Connections

Table 3. Connection Requirements

Connection	Requirement	Consequence if omitted
`s_axi` to the master	Required	Core does nothing
`m_axi` to the peripheral	Required	Core does nothing
`s_axi_ctrl` to a master	Required	Rule table cannot be programmed; core denies all traffic
`irq` to a CPU interrupt	Recommended	Faults must be polled instead
Software-controllable reset on the protected peripheral	**Required for recovery**	Cannot recover from a timeout without a full system reset

> **Caution**
> The core does not drive the protected peripheral's reset. It did in version 1.x. Recovery from a timeout still requires that the peripheral be reset, so the reset must be reachable from software — a Platform Designer reset bridge under software control, or a PIO output, both work. Without it a timeout is unrecoverable short of resetting the whole system. See Section 3.5.

2.4 Simulating the IP Core

Three flows are supported. All three run the same testbench.

Verilator — no licence required, recommended for continuous integration:

```
simulation/verilator/run_sim.sh
```

The script runs a strict elaboration check, lints the RTL, builds, and runs the regression. Its exit status is 0 only if every check passes.

Questa — adds functional coverage and assertion non-vacuity reporting:

```
cd simulation/questa
vsim -c -do run_sim.tcl
```

Writes `coverage.ucdb`, `coverage_report.txt` and `run.log`. The assertion and cover-directive results are inside `coverage_report.txt`, under the bound SVA instance.

Icarus Verilog — functional tests only; the assertion bind is skipped:

```
iverilog -g2012 -DICARUS -o tb.out \
    rtl/axi_firewall_regs.sv rtl/axi_firewall_top.sv tb/axi_firewall_tb.sv
vvp tb.out
```

> **Note**
> Use `-g2012`. The sources are SystemVerilog.

2.5 Files Provided

Table 4. Files Provided

File	Description
`rtl/axi_firewall_top.sv`	Top level: datapaths, timeout, isolation, block/unblock
`rtl/axi_firewall_regs.sv`	Register block: rule table, status, interrupt, control slave
`axi_firewall_hw.tcl`	Platform Designer component description
`tb/axi_firewall_tb.sv`	Self-checking testbench, 103 checks
`tb/axi_firewall_sva.sv`	SystemVerilog assertions and cover points
`verification/orphan_response_tb.sv`	Standalone measurement of the recovery hazard
`simulation/questa/run_sim.tcl`	Questa flow with coverage
`simulation/verilator/run_sim.sh`	Verilator flow
`simulation/verilator/slangcheck.py`	Strict elaboration gate
`doc/`	Block diagrams and this user guide

3. Functional Description

3.1 Block Diagram

The core presents three AXI4-Lite interfaces and one interrupt output.

Traffic flows from the master into `s_axi`, is checked, and is either forwarded to the peripheral through `m_axi` or answered locally with an error response. Configuration and status are reached through `s_axi_ctrl`, which is independent of the data path — so configuring or inspecting the core is never subject to a firewall rule, and never blockable by a peripheral that has stopped answering. Figure 1 shows how the core wires into a system.

Internally the core contains two independent datapaths — one for writes, one for reads — and a shared register block. Each datapath has its own state machine, its own capture registers, its own rule-lookup port and its own fault signals, so a read and a write may be in flight simultaneously. The only shared state is the fault capture pair `FAULT_ADDR` and `FAULT_INFO`. Figure 2 shows the internal structure.

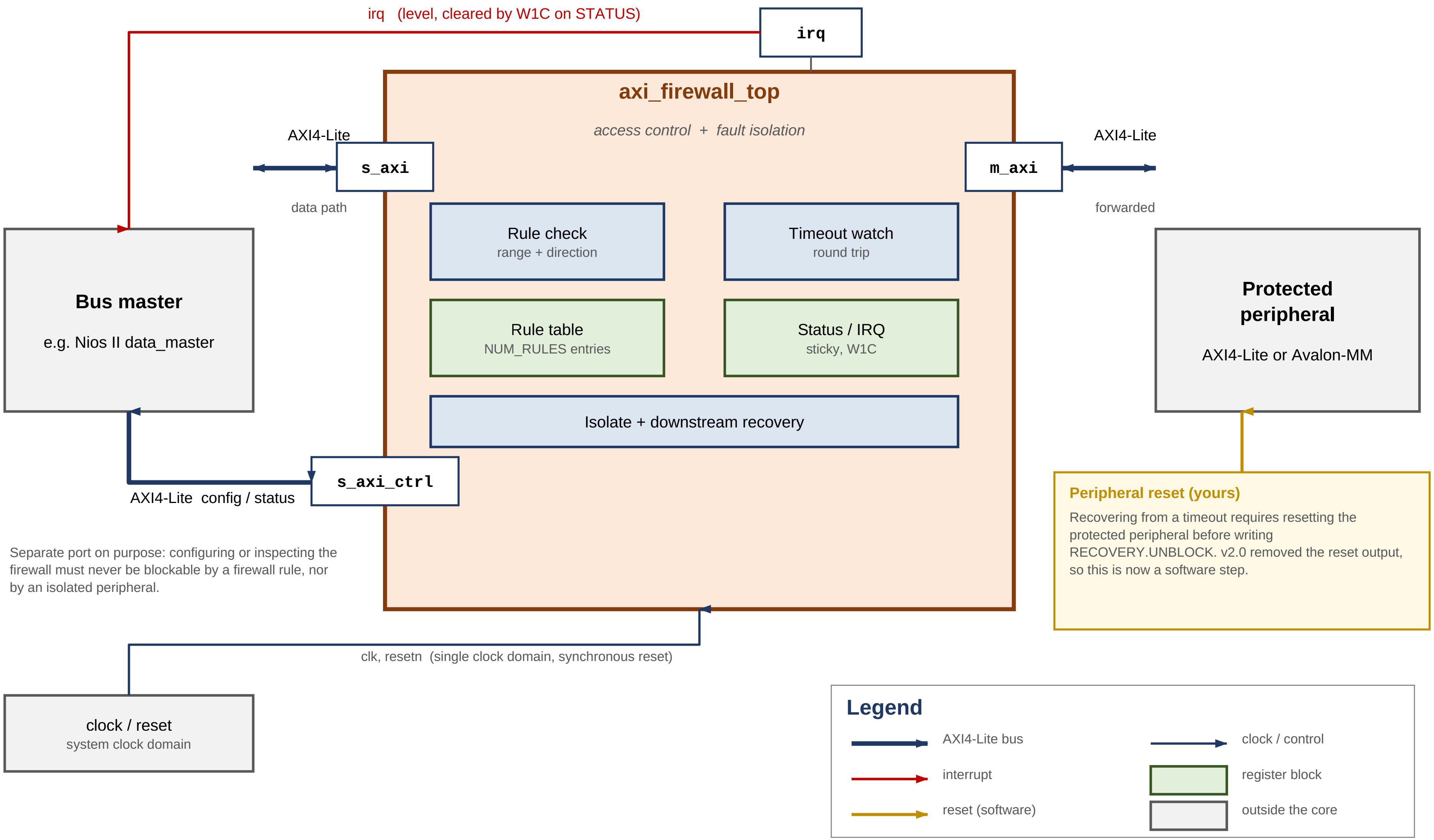


Figure 1. System Context

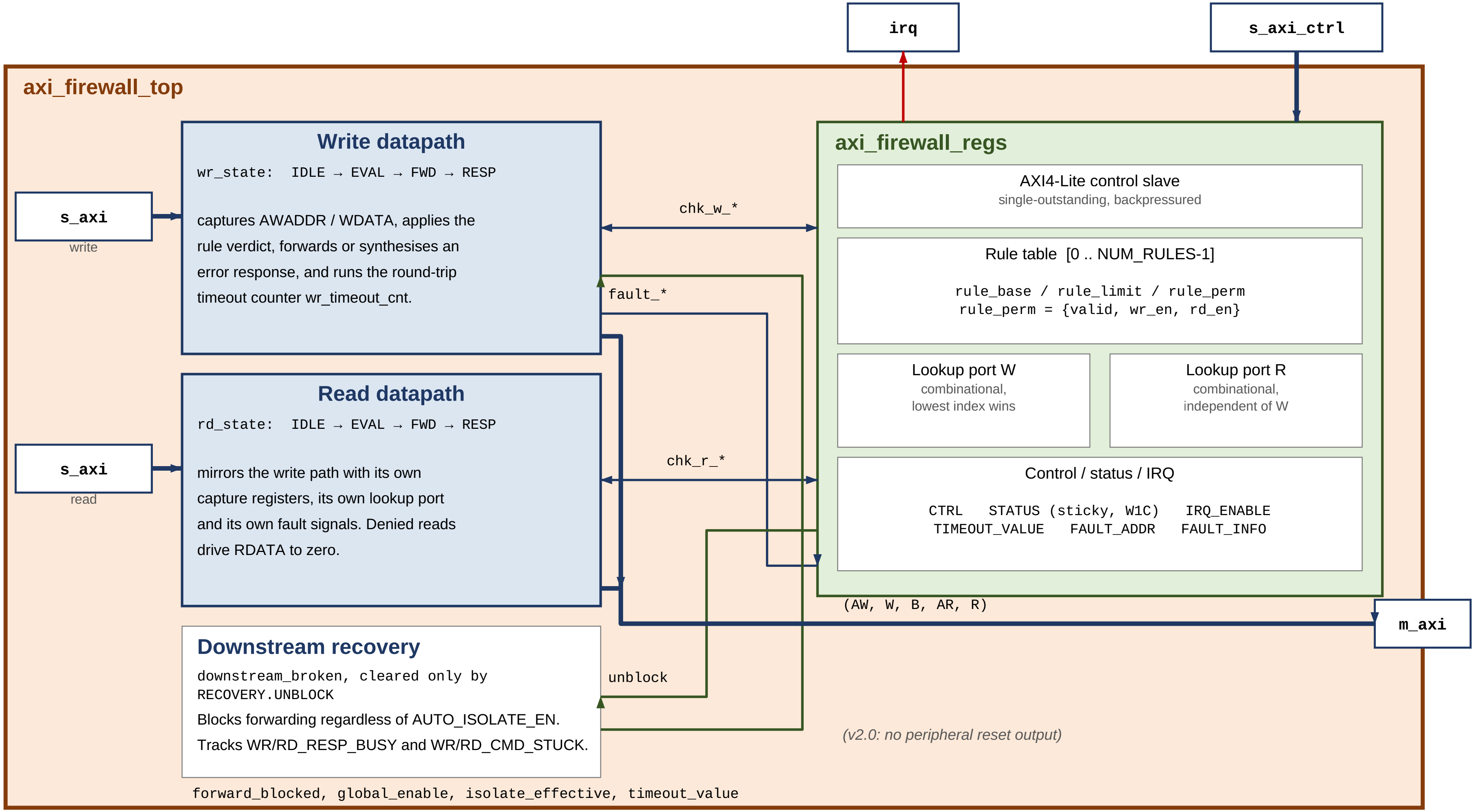


Figure 2. Internal Architecture

3.2 Access Control

Every transaction's address is compared against the rule table. Each rule is an inclusive address range with independent read and write permission, and a valid bit.

The verdict is applied in the EVAL state, in strict priority order:

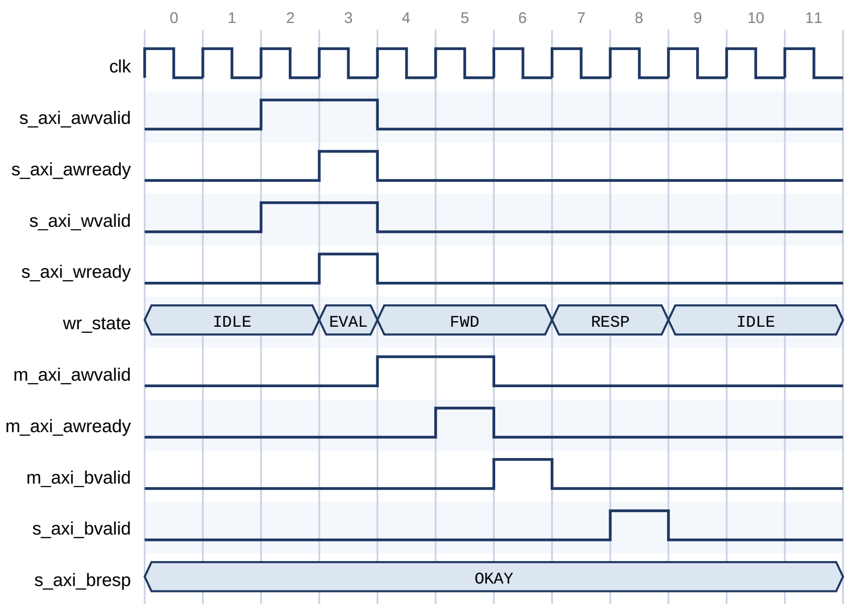
1. If the downstream is blocked or the core is isolated, respond SLVERR.
2. If `CTRL.GLOBAL_ENABLE` is clear, forward unconditionally (bypass mode).
3. If no valid rule contains the address, respond DECERR.
4. If a rule contains the address but does not permit this direction, respond SLVERR.
5. Otherwise, forward the transaction.

Table 5. Response Encoding

Condition	Response	Status bit set	Interrupt
Permitted, peripheral answers	As returned by the peripheral	—	No
No valid rule matches	`DECERR`	`ADDR_VIOLATION`	Yes, if enabled
Rule matches, direction denied	`SLVERR`	`PERM_VIOLATION`	Yes, if enabled
Blocked or isolated	`SLVERR`	None	No
Peripheral timed out	`SLVERR`	`TIMEOUT_ERROR`	Yes, if enabled

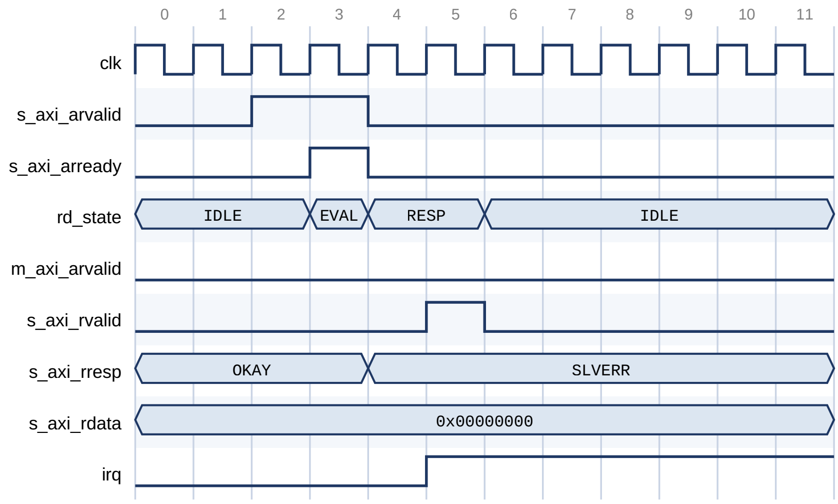
A transaction rejected because the core is isolated or blocked sets no status bit and raises no interrupt. This is deliberate: only genuine policy violations and timeouts are logged, so a burst of rejected traffic during isolation cannot bury the fault that caused it.

A denied read drives `RDATA` to zero rather than leaving the previous value on the bus, so a rejected read cannot leak the result of an earlier permitted one.



The address matches a valid rule that permits writes, so the transaction is forwarded on m_axi.
Cycle 1: request accepted (AWREADY and WREADY together). Cycle 6: BVALID returns OKAY.

Figure 3. Permitted Write — Request to Response, 6 Cycles



The address falls in a write-only rule. RD_EVAL rejects it and answers SLVERR from RD_RESP.
m_axi_arvalid never asserts - the peripheral is not touched. RDATA is driven to zero, not left stale.
irq asserts because STATUS.PERM_VIOLATION is sticky and its interrupt is enabled.

Figure 4. Permission-Denied Read — Answered Locally, Nothing Reaches the Peripheral

In Figure 4, note that `m_axi_arvalid` never asserts. The peripheral is not touched at all, `RDATA` is driven to zero, and `irq` asserts because the sticky `PERM_VIOLATION` bit was set with its interrupt enabled.

3.3 Rule Evaluation

The lookup is a combinational priority chain: the **lowest-index valid rule** containing the address wins. Ranges need not be disjoint — if you use overlapping ranges, place the more specific rule at the lower index.

Two independent lookup ports are provided, one per datapath, so a read and a write are resolved in the same cycle without contending for a shared comparator.

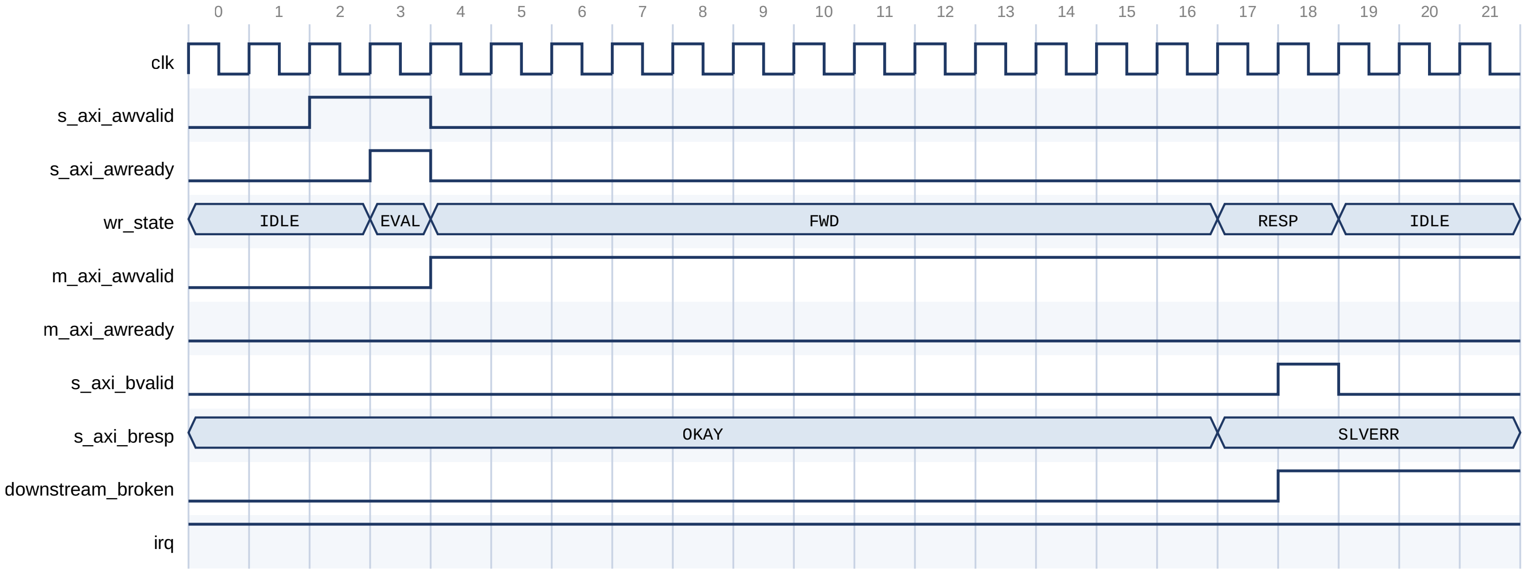
The lookup is purely combinational, so the verdict costs no extra cycle: EVAL is always one cycle.

3.4 Fault Isolation and Timeout

Every forwarded transaction is watched by a counter loaded from `TIMEOUT_VALUE`. The counter covers the **entire round trip** — from the moment forwarding begins to the arrival of the response — so it catches both a peripheral that never accepts the command and one that accepts it and then goes quiet.

On expiry the core:

1. answers **SLVERR upstream immediately**, so the master never hangs;
2. latches an internal *downstream-blocked* state that blocks all further forwarding, independently of `CTRL.AUTO_ISOLATE_EN`;
3. **leaves any asserted `m_axi_*VALID` asserted**, because AXI requires VALID to remain asserted until READY.



The peripheral never raises AWREADY. TIMEOUT_VALUE is 12 clocks in this capture.
On expiry the core answers SLVERR upstream so the master never hangs, and latches downstream_broken.
m_axi_awvalid stays asserted: AXI forbids withdrawing VALID before READY. Only UNBLOCK may drop it.

Figure 5. Downstream Timeout — Master Answered Immediately, VALID Left Asserted

Caution

Withdrawing an asserted VALID before its handshake is an AXI protocol violation that can wedge the interconnect between the core and the peripheral, not merely the peripheral. The core therefore never does so on timeout. The stuck VALID is withdrawn at exactly one point — the `RECOVERY.UNBLOCK` write described in Section 3.5.

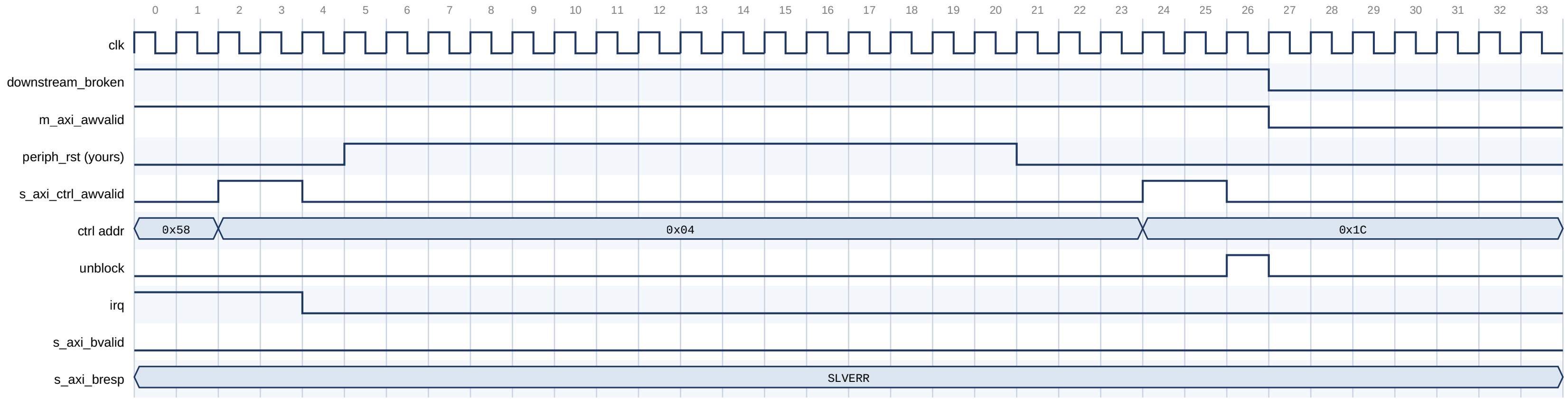
`CTRL.AUTO_ISOLATE_EN` governs only whether the visible `ISOLATED` status bit is set. Blocking after a timeout is required for protocol safety and happens either way.

3.5 Recovery Sequence

Recovery is an explicit software sequence. It is modelled on the procedure AMD document for their AXI Firewall, which carries the same requirement to reset the monitored side before unblocking.

Table 6. Recovery Sequence

Step	Action	Why
1	Stop issuing transactions to `s_axi`	New transactions are rejected while blocked
2	Write 1 to the sticky `STATUS` bits	Acknowledges the fault and releases the auto-isolate latch. **Required before step 5** — see below
3	Poll `STATUS` until `WR_RESP_BUSY` and `RD_RESP_BUSY` clear — **with a bound**	Tells you no late response is still in flight
4	**Reset the protected peripheral** (≥ 16 clocks)	Discards the peripheral's AXI state
5	Write 1 to `RECOVERY.UNBLOCK`	Reopens forwarding and withdraws the stuck VALID
6	Resume, retrying anything that failed	Transactions attempted while blocked returned SLVERR



Write 0x04 acknowledges the fault (irq drops). The peripheral reset is driven by the system, not the core.
Write 0x1C pulses unblock: downstream_broken clears and the stuck m_axi_awvalid is withdrawn in the same event.
The final write completes with OKAY, showing forwarding has reopened.

Figure 6. Recovery Sequence — Acknowledge, Reset the Peripheral, UNBLOCK

Caution

Step 4 is not optional. `UNBLOCK` causes the core to withdraw an asserted VALID. If the peripheral has not been reset, that is a protocol violation on a live bus, and the peripheral may additionally still act on a transaction the core already reported to the master as failed. Measured: 0 of 25 tested timing offsets mis-attribute a stale response when the sequence is followed; 1 of 25 does when step 4 is skipped.

Caution

Step 2 must precede step 5, and the order is load-bearing. Forwarding is gated by *two* independent conditions — the blocked state and the isolate state — and each is cleared by a different write. `UNBLOCK` clears only the blocked state. If `AUTO_ISOLATE_EN` was set when the timeout occurred, the auto-isolate latch is also set, and only the W1C in step 2 releases it. Writing `UNBLOCK` without having acknowledged `STATUS` first leaves the core isolated, and every transaction still returns SLVERR. `STATUS` then reads `BLOCKED = 0` with `ISOLATED = 1`, which is the signature of this mistake.

3.5.1 Bounding the Busy Poll

`WR_RESP_BUSY` and `RD_RESP_BUSY` mean *the peripheral has accepted a command and still owes a response*. A peripheral that accepted a command and then died owes one forever, so an unbounded poll in step 3 hangs precisely when recovery matters most.

Treat the bits as advisory:

- **Clear** — no late response can still be in flight; the reset is unambiguously safe.
- **Stuck** — reset anyway, and let `UNBLOCK` discard what is owed.

`WR_CMD_STUCK` and `RD_CMD_STUCK` report the other case: a command the peripheral never accepted at all. These can only be cleared by `UNBLOCK`, so a driver that sees one knows immediately that polling `RESP_BUSY` will not be enough.

Note

This differs from AMD's core, which autonomously flushes outstanding transactions when blocked so that its busy bits always reach zero. Removing this caveat by adding autonomous flushing is the most valuable planned enhancement to this core.

3.6 Interrupts

`irq` is a level interrupt, asserted while any sticky `STATUS` bit is set and its corresponding `IRQ_ENABLE` bit is set:

```
irq = (ADDR_VIOLATION  & IRQ_ENABLE[0])
    | (PERM_VIOLATION  & IRQ_ENABLE[1])
    | (TIMEOUT_ERROR   & IRQ_ENABLE[2])
```

Clear the interrupt at its source by writing 1 to the relevant `STATUS` bit. There is no separate acknowledge register. This is the standard memory-mapped-peripheral idiom and works directly with the Nios II HAL ISR pattern.

`FAULT_ADDR` and `FAULT_INFO` capture the address and type of the most recent fault of any kind.

Caution

If the read and write datapaths fault in the *same cycle* with *different* fault types, the capture registers can disagree with each other. `FAULT_ADDR` and `FAULT_INFO.WAS_WRITE` always take the write side. `FAULT_INFO.FAULT_TYPE` is resolved separately, by type precedence across both datapaths — `TIMEOUT` beats `PERM` beats `ADDR`. So a simultaneous write-`ADDR` and read-`PERM` fault reports `WAS_WRITE = 1` with the write's address, but `FAULT_TYPE = PERM`, which belongs to the read.

The sticky `STATUS` bits are unaffected — every fault sets its own bit correctly. Only the single shared capture pair is ambiguous. Software that must attribute every fault individually should correlate with its own transaction log rather than rely on `FAULT_INFO` alone; see Section 9.7.

3.7 Control Port Behaviour

`s_axi_ctrl` is single-outstanding and applies backpressure accordingly: `AWREADY` and `WREADY` are withheld while a `BVALID` is unacknowledged, and `ARREADY` while an `RVALID` is. A master that issues one access at a time sees no difference; one that pipelines simply waits.

Writes to reserved or unaligned offsets are ignored and answered `OKAY`; reads of them return zero.

The control port is deliberately independent of the data path. Configuring or inspecting the core is never subject to a firewall rule, and never blockable by an isolated or hung peripheral — which is what makes recovery possible at all.

3.8 Reset

`resetn` is active-low and **synchronous**. It resets the entire core: both datapath state machines, the rule table, all status and configuration registers, and the blocked state.

A global reset is therefore also the escape hatch of last resort for a downstream that never recovers — it clears the blocked state without an `UNBLOCK`, at the cost of losing the rule table, which must be reprogrammed.

Out of reset the core is **secure by default**: `GLOBAL_ENABLE` is 1 and the rule table is empty, so every transaction is denied until rules are programmed.

3.9 Latency

Table 7. Latency, Zero-Wait-State Peripheral

Operation	Cycles
Single write, request asserted to `BVALID`	6
Single read, request asserted to `RVALID`	6

Measured in simulation against a zero-wait-state peripheral model, counting clock edges from request assertion to response valid. The testbench fails the run if either exceeds 8 cycles, so the figures cannot silently regress.

The core is single-outstanding and non-pipelined, so this cost is per transaction and does not amortise.

4. Parameters

All five parameters are compile-time. Nothing in the core is runtime-reconfigurable except the rule table and the control registers.

Table 8. Parameters

Parameter	Type	Default	Legal range	Description
`ADDR_WIDTH`	Integer	32	8–32	Width of `s_axi` / `m_axi` address buses. Also the width of every rule base and limit, and of `FAULT_ADDR`.
`DATA_WIDTH`	Integer	32	32, 64	Width of `s_axi` / `m_axi` data buses. `WSTRB` is `DATA_WIDTH/8` bits.
`CTRL_ADDR_WIDTH`	Integer	12	8–16	Width of the `s_axi_ctrl` address bus.
`NUM_RULES`	Integer	8	1–64	Number of address-range rules.
`TIMEOUT_WIDTH`	Integer	20	8–32	Width of the timeout counter. Maximum programmable timeout is $2^{\mathtt{TIMEOUT\_WIDTH}} - 1$ clock cycles.

Ranges are enforced by `axi_firewall_hw.tcl`. `DATA_WIDTH` is restricted to 32 and 64 because AXI4-Lite defines only those two.

Sizing `CTRL_ADDR_WIDTH`

The register file spans `0x40 + NUM_RULES × 16` bytes, so:

```
CTRL_ADDR_WIDTH ≥ ceil(log2(0x40 + NUM_RULES × 16))
```

Table 9. Minimum `CTRL_ADDR_WIDTH`

`NUM_RULES`	Table spans	Highest byte used	Minimum `CTRL_ADDR_WIDTH`
1	0x50 bytes	0x4F	7
4	0x80 bytes	0x7F	7
8 (default)	0xC0 bytes	0xBF	8
16	0x140 bytes	0x13F	9
32	0x240 bytes	0x23F	10
64	0x440 bytes	0x43F	11

The default of 12 covers every legal `NUM_RULES`. A validation callback in `axi_firewall_hw.tcl` rejects an undersized value, because the failure mode is silent: high-index rules simply become unreachable, the rule table looks like it programmed correctly, and the firewall quietly enforces fewer rules than you configured.

Sizing `TIMEOUT_WIDTH`

Choose the timeout to be comfortably longer than the protected peripheral's worst-case response, including any interconnect arbitration. `TIMEOUT_WIDTH` only has to be wide enough to express it.

Table 10. Maximum Timeout by Width

`TIMEOUT_WIDTH`	Maximum count	At 100 MHz	At 50 MHz
8	255	2.6 µs	5.1 µs
12	4 095	41 µs	82 µs
16	65 535	655 µs	1.3 ms
20 (default)	1 048 575	10.5 ms	21 ms
24	16 777 215	168 ms	336 ms
32	4 294 967 295	43 s	86 s

> **Note**
>
> `TIMEOUT_VALUE` resets to all ones, which at the default width is 10.5 ms at 100 MHz — long enough that the timeout is effectively disabled until software programs a real value. This is deliberate: the reset state denies traffic on policy grounds, but never manufactures a spurious timeout during boot.

5. Interface Signals

The core has 60 ports across six Platform Designer interfaces.

Table 11. Interfaces

Interface	Type	Role	Signals
`clock`	Clock sink	Single clock domain	1
`reset`	Reset sink	Active-low, synchronous	1
`s_axi`	AXI4-Lite slave	Protected data path, master side	19
`m_axi`	AXI4-Lite master	Protected data path, peripheral side	19
`s_axi_ctrl`	AXI4-Lite slave	Configuration and status	19
`irq`	Interrupt sender	Fault interrupt	1

5.1 Clock and Reset

Table 12. Clock and Reset Signals

Signal	Direction	Width	Description
`clk`	Input	1	Clock. All three interfaces are synchronous to it; there is no CDC in the core.
`resetn`	Input	1	Active-low **synchronous** reset. Resets the FSMs, rule table and all registers.

5.2 `s_axi` — Protected Slave

Connect to the bus master. Standard AXI4-Lite slave.

Table 13. `s_axi` **Signals**

Signal	Direction	Width	Description
`s_axi_awaddr`	Input	`ADDR_WIDTH`	Write address
`s_axi_awprot`	Input	3	Protection type. Accepted and forwarded unchanged; **not** used in rule evaluation.
`s_axi_awvalid`	Input	1	Write address valid
`s_axi_awready`	Output	1	Write address ready
`s_axi_wdata`	Input	`DATA_WIDTH`	Write data
`s_axi_wstrb`	Input	`DATA_WIDTH/8`	Write strobes
`s_axi_wvalid`	Input	1	Write data valid
`s_axi_wready`	Output	1	Write data ready
`s_axi_bresp`	Output	2	Write response: `OKAY` , `SLVERR` or `DECERR`
`s_axi_bvalid`	Output	1	Write response valid
`s_axi_bready`	Input	1	Write response ready
`s_axi_araddr`	Input	`ADDR_WIDTH`	Read address
`s_axi_arprot`	Input	3	Protection type. Forwarded unchanged; not used in rule evaluation.
`s_axi_arvalid`	Input	1	Read address valid
`s_axi_arready`	Output	1	Read address ready
`s_axi_rdata`	Output	`DATA_WIDTH`	Read data. Driven to zero on any denied read.
`s_axi_rresp`	Output	2	Read response
`s_axi_rvalid`	Output	1	Read data valid
`s_axi_rready`	Input	1	Read data ready

> **Note**
> `AWPROT` / `ARPROT` are transported but not policed. The rule table keys on address and direction only. If you need privilege-based filtering, that is a feature addition, not a configuration option.

5.3 `m_axi` — Protected Master

Connect to the peripheral being protected. Signal names and widths mirror `s_axi` with directions reversed.

Table 14. `m_axi` **Signals of Note**

Signal	Direction	Width	Description
`m_axi_awaddr`	Output	`ADDR_WIDTH`	Forwarded write address, unmodified
`m_axi_awvalid`	Output	1	Held asserted until `AWREADY` , including across a timeout
`m_axi_wvalid`	Output	1	Held asserted until `WREADY` , including across a timeout
`m_axi_bready`	Output	1	**Tied high.** The core always accepts a write response.
`m_axi_rready`	Output	1	**Tied high.** The core always accepts read data.

Caution

`m_axi_bready` and `m_axi_rready` are tied high so that a peripheral can never be back-pressured into a deadlock by the firewall itself. The consequence is that a late response from an abandoned transaction is accepted rather than rejected, which is exactly why the recovery sequence resets the peripheral before unblocking. See Section 3.5.

5.4 `s_axi_ctrl` — Control and Status Slave

A separate AXI4-Lite slave carrying the register file. `WDATA` is always 32 bits wide and `WSTRB` always 4 bits, independent of `DATA_WIDTH`; the addressable range is set by `CTRL_ADDR_WIDTH`.

Table 15. `s_axi_ctrl` **Signals of Note**

Signal	Direction	Width	Description
`s_axi_ctrl_awaddr`	Input	`CTRL_ADDR_WIDTH`	Register byte offset
`s_axi_ctrl_wdata`	Input	32	Always 32 bits
`s_axi_ctrl_wstrb`	Input	4	Byte strobes; honoured for partial-width writes
`s_axi_ctrl_bresp`	Output	2	Always `OKAY`; writes to reserved offsets are ignored, not errored
`s_axi_ctrl_rdata`	Output	32	Reserved offsets read as zero

5.5 `irq`

Table 16. `irq`

Signal	Direction	Width	Description
`irq`	Output	1	Active-high **level** interrupt. Asserted while any enabled sticky `STATUS` bit is set. Cleared by W1C on `STATUS`.

6. Register Map

All registers are 32 bits and word-aligned on `s_axi_ctrl`. Offsets are byte offsets from the base address assigned to `s_axi_ctrl`.

Table 17. Register Map

Offset	Name	Access	Reset	Description
0x00	`CTRL`	R/W	0x3	Global enable, auto-isolate, manual isolate
0x04	`STATUS`	R / W1C	0x0	Sticky faults, live isolation and outstanding status
0x08	`IRQ_ENABLE`	R/W	0x7	Per-fault interrupt enables
0x0C	`TIMEOUT_VALUE`	R/W	all ones	Round-trip timeout, in clock cycles
0x10	`FAULT_ADDR`	R	0x0	Address of the most recent fault
0x14	`FAULT_INFO`	R	0x0	Type and direction of the most recent fault
0x18	`CORE_INFO`	R	—	Version and `NUM_RULES`
0x1C	`RECOVERY`	W	—	`UNBLOCK`. Reads as zero.
0x20–0x3F	*reserved*	—	—	Writes ignored, reads return zero
0x40 + i·0x10	`RULE_BASE[i]`	R/W	0x0	Inclusive range base
0x44 + i·0x10	`RULE_LIMIT[i]`	R/W	0x0	Inclusive range limit
0x48 + i·0x10	`RULE_PERM[i]`	R/W	0x0	Permissions and valid bit
0x4C + i·0x10	*reserved*	—	—	Padding to a 16-byte stride

`i` runs 0 to `NUM_RULES - 1`. With the default 8 rules the table spans 0x40–0xBF.

6.1 CTRL (0x00)

Table 18. CTRL

Bit	Name	Access	Reset	Description
31:3	reserved	R	0	Reads zero
2	`MANUAL_ISOLATE`	R/W	0	1 = reject all traffic with `SLVERR`, regardless of the rule table
1	`AUTO_ISOLATE_EN`	R/W	1	1 = a timeout also sets the visible `ISOLATED` status
0	`GLOBAL_ENABLE`	R/W	1	1 = enforce the rule table; 0 = forward everything unchecked

> **Caution**
> Clearing `GLOBAL_ENABLE` disables access control entirely — the core becomes a transparent pass-through with only the timeout still active. It is a bring-up and debug aid, not an operating mode.

6.2 STATUS (0x04)

Table 19. STATUS

Bit	Name	Access	Reset	Description
31:9	reserved	R	0	Reads zero
8	`RD_CMD_STUCK`	R	0	Blocked **and** `m_axi_arvalid` still asserted — a read the peripheral never accepted
7	`WR_CMD_STUCK`	R	0	Blocked **and** `m_axi_awvalid` or `m_axi_wvalid` still asserted — a write the peripheral never accepted
6	`RD_RESP_BUSY`	R	0	The peripheral accepted a read command and still owes `RVALID`
5	`WR_RESP_BUSY`	R	0	The peripheral accepted a full write and still owes `BVALID`
4	`BLOCKED`	R	0	Forwarding is blocked after a timeout; cleared only by `RECOVERY.UNBLOCK`
3	`ISOLATED`	R	0	`MANUAL_ISOLATE` or the auto-isolate latch is active
2	`TIMEOUT_ERROR`	R/W1C	0	Sticky: a forwarded transaction timed out
1	`PERM_VIOLATION`	R/W1C	0	Sticky: a rule matched but denied the direction
0	`ADDR_VIOLATION`	R/W1C	0	Sticky: no valid rule matched the address

Bits 8:3 are live status, not sticky, and cannot be written. Bits 2:0 are write-1-to-clear; writing 0 to a bit leaves it unchanged.

> **Caution**
>
> Writing 1 to `TIMEOUT_ERROR` acknowledges the fault and releases the auto-isolate latch, but does **not** clear `BLOCKED` and does not resume forwarding. That requires `RECOVERY.UNBLOCK`. In version 1.x the W1C alone resumed traffic; the separation exists so that acknowledging a fault cannot accidentally restart traffic toward a peripheral nobody has reset.

6.3 IRQ_ENABLE (0x08)

Table 20. IRQ_ENABLE

Bit	Name	Access	Reset	Description
31:3	reserved	R	0	Reads zero
2	`EN_TIMEOUT`	R/W	1	Enable `irq` on `TIMEOUT_ERROR`
1	`EN_PERM`	R/W	1	Enable `irq` on `PERM_VIOLATION`
0	`EN_ADDR`	R/W	1	Enable `irq` on `ADDR_VIOLATION`

Masking a bit suppresses the interrupt only. The corresponding `STATUS` bit is still set and still requires a W1C to clear.

6.4 TIMEOUT_VALUE (0x0C)

Table 21. TIMEOUT_VALUE

Bit	Name	Access	Reset	Description
`31:TIMEOUT_WIDTH`	reserved	R	0	Reads zero; writes to these bits are discarded
`TIMEOUT_WIDTH-1:0`	`TIMEOUT`	R/W	all ones	Round-trip timeout in clock cycles

Byte strobes are honoured, so a byte- or halfword-wide write updates only the addressed bytes of the field. Programming 0 makes every forwarded transaction time out immediately; the core does not reject the value.

6.5 FAULT_ADDR (0x10)

Table 22. FAULT_ADDR

Bit	Name	Access	Reset	Description
`31:ADDR_WIDTH`	reserved	R	0	Reads zero when `ADDR_WIDTH` < 32
`ADDR_WIDTH-1:0`	`ADDR`	R	0	Address of the most recent fault

Overwritten by every new fault, including one whose interrupt is masked. Read it before clearing `STATUS` if you intend to log it.

6.6 FAULT_INFO (0x14)

Table 23. FAULT_INFO

Bit	Name	Access	Reset	Description
31:4	reserved	R	0	Reads zero
3:1	`FAULT_TYPE`	R	0	See table below
0	`WAS_WRITE`	R	0	1 = the faulting transaction was a write

Table 24. FAULT_TYPE Encoding

Value	Name	Meaning
0	`NONE`	No fault recorded since reset
1	`ADDR`	No valid rule matched — answered `DECERR`
2	`PERM`	A rule matched but denied the direction — answered `SLVERR`
3	`TIMEOUT`	The peripheral did not complete in time — answered `SLVERR`

Values 4–7 are not generated.

6.7 CORE_INFO (0x18)

Table 25. CORE_INFO

Bit	Name	Access	Description
31:16	`VERSION`	R	0x0200 for version 2.0
15:8	reserved	R	Reads zero
7:0	`NUM_RULES`	R	Rule count this instance was built with

Read `NUM_RULES` at run time rather than hard-coding it, so one driver binary works across instances. Check `VERSION` before running the recovery sequence: 1.x and 2.0 recover differently.

6.8 RECOVERY (0x1C)

Table 26. RECOVERY

Bit	Name	Access	Description
31:1	reserved	W	Ignored
0	`UNBLOCK`	W1S, self-clearing	Writing 1 clears `BLOCKED`, reopens forwarding and withdraws any stuck `m_axi VALID`

Reads return zero. Writing 0 does nothing. New in version 2.0.

> **Caution**
> Only write `UNBLOCK` after resetting the protected peripheral. This is the one point at which the core withdraws an asserted `VALID`, which is safe against a peripheral that has just been reset and a protocol violation against one that has not.

6.9 Rule Table

Each rule occupies a 16-byte slot at `0x40 + i × 0x10`.

Table 27. Rule Slot Layout

Sub-offset	Name	Access	Reset	Description
+0x0	`RULE_BASE[i]`	R/W	0	Inclusive base address
+0x4	`RULE_LIMIT[i]`	R/W	0	Inclusive limit address
+0x8	`RULE_PERM[i]`	R/W	0	Permissions, see below
+0xC	reserved	—	—	Padding

Table 28. RULE_PERM[i]

Bit	Name	Access	Reset	Description
31:3	reserved	R	0	Reads zero
2	`VALID`	R/W	0	1 = this rule takes part in matching
1	`WRITE_ALLOW`	R/W	0	1 = permit writes in this range
0	`READ_ALLOW`	R/W	0	1 = permit reads in this range

A rule matches when `VALID` is set and `RULE_BASE[i] ≤ address ≤ RULE_LIMIT[i]`. Both bounds are inclusive, so a single-word window is `BASE == LIMIT`. `BASE > LIMIT` matches nothing; the core does not reject it.

The lowest-index matching valid rule wins. Only its permission bits are consulted — a later, more permissive rule covering the same address has no effect.

`RULE_BASE` and `RULE_LIMIT` honour byte strobes, so a byte-wide write updates only the addressed bytes. `RULE_PERM` is updated only when `WSTRB[0]` is set.

7. Software Programming Model

Register offsets in the examples below are relative to the base address assigned to `s_axi_ctrl`.

```
#define FW_CTRL          0x00
#define FW_STATUS        0x04
#define FW_IRQ_ENABLE    0x08
#define FW_TIMEOUT       0x0C
#define FW_FAULT_ADDR    0x10
#define FW_FAULT_INFO    0x14
#define FW_CORE_INFO     0x18
#define FW_RECOVERY      0x1C
#define FW_RULE(i)       (0x40 + (i) * 0x10)
#define FW_RULE_BASE(i)  (FW_RULE(i) + 0x0)
#define FW_RULE_LIMIT(i) (FW_RULE(i) + 0x4)
#define FW_RULE_PERM(i)  (FW_RULE(i) + 0x8)

#define FW_PERM_VALID    (1u << 2)
#define FW_PERM_WRITE    (1u << 1)
#define FW_PERM_READ     (1u << 0)

#define FW_ST_ADDR_VIOL  (1u << 0)
#define FW_ST_PERM_VIOL  (1u << 1)
#define FW_ST_TIMEOUT    (1u << 2)
#define FW_ST_ISOLATED   (1u << 3)
#define FW_ST_BLOCKED    (1u << 4)
#define FW_ST_WR_BUSY    (1u << 5)
#define FW_ST_RD_BUSY    (1u << 6)
#define FW_ST_WR_STUCK   (1u << 7)
#define FW_ST_RD_STUCK   (1u << 8)
#define FW_ST_FAULTS     (FW_ST_ADDR_VIOL | FW_ST_PERM_VIOL | FW_ST_TIMEOUT)
```

7.1 Initialisation

Program the rule table before allowing any traffic to reach `s_axi`. Out of reset the table is empty and `GLOBAL_ENABLE` is set, so every transaction is denied until you configure it.

```c
void fw_init(void *base, uint32_t timeout_cycles)
{
    /* Rules are only meaningful up to the count this instance was built
       with. Read it rather than assuming the default. */
    uint32_t n_rules = IORD_32DIRECT(base, FW_CORE_INFO) & 0xFF;

    for (uint32_t i = 0; i < n_rules; i++)
        IOWR_32DIRECT(base, FW_RULE_PERM(i), 0);   /* invalidate */

    IOWR_32DIRECT(base, FW_TIMEOUT, timeout_cycles);
    IOWR_32DIRECT(base, FW_STATUS, FW_ST_FAULTS);  /* clear stale sticky */
    IOWR_32DIRECT(base, FW_IRQ_ENABLE, 0x7);
}

void fw_add_rule(void *base, uint32_t i,
                 uint32_t lo, uint32_t hi, int can_read, int can_write)
{
    /* Program the range before the valid bit. Setting VALID first would
       expose a window in which the rule is live over a stale range. */
    IOWR_32DIRECT(base, FW_RULE_BASE(i),  lo);
    IOWR_32DIRECT(base, FW_RULE_LIMIT(i), hi);
    IOWR_32DIRECT(base, FW_RULE_PERM(i),
                  FW_PERM_VALID
                  | (can_write ? FW_PERM_WRITE : 0)
                  | (can_read  ? FW_PERM_READ  : 0));
}
```

Caution

Write `RULE_PERM` last. The rule table is live: a rule becomes active the instant its `VALID` bit is set, and a transaction arriving in the window between setting `VALID` and updating the range would be judged against whatever the range registers happened to hold.

7.2 Changing a Rule at Run Time

To narrow a rule safely, clear `VALID` first, change the range, then set `VALID` again. Widening a rule can be done range-first without an intermediate invalidation, because the intermediate state is never more permissive than either endpoint.

7.3 Interrupt Service Routine

```
void fw_isr(void *base)
{
    uint32_t st = IORD_32DIRECT(base, FW_STATUS);

    /* Capture before acknowledging: a later fault overwrites these. */
    uint32_t addr = IORD_32DIRECT(base, FW_FAULT_ADDR);
    uint32_t info = IORD_32DIRECT(base, FW_FAULT_INFO);

    if (st & FW_ST_ADDR_VIOL) log_addr_violation(addr, info & 1);
    if (st & FW_ST_PERM_VIOL) log_perm_violation(addr, info & 1);
    if (st & FW_ST_TIMEOUT)   schedule_recovery();

    /* W1C only the bits actually seen, so a fault raised between the read
       and the write is not silently discarded. */
    IOWR_32DIRECT(base, FW_STATUS, st & FW_ST_FAULTS);
}
```

Do not run the recovery sequence from inside the ISR. It resets the peripheral and polls, which does not belong in interrupt context.

7.4 Recovery from a Timeout

This implements the sequence in Section 3.5.

```
int fw_recover(void *base, void (*reset_peripheral)(void))
{
    /* Step 1 is the caller's: stop issuing transactions to the
       protected peripheral before calling this function. */

    /* Step 2: acknowledge the fault and release the auto-isolate latch. */
    IOWR_32DIRECT(base, FW_STATUS, FW_ST_FAULTS);

    /* Step 3: bounded poll. A peripheral that accepted a command and then
       died owes a response forever, so this MUST NOT be unbounded. On
       expiry, continue anyway - UNBLOCK discards what is still owed. */
    for (int spin = 0; spin < 1000; spin++) {
        uint32_t st = IORD_32DIRECT(base, FW_STATUS);
        if (!(st & (FW_ST_WR_BUSY | FW_ST_RD_BUSY)))
            break;
    }

    /* Step 4: reset the protected peripheral. Not optional. Hold for at
       least 16 clocks so its AXI state machines fully re-initialise. */
    reset_peripheral();

    /* Step 5: reopen the downstream. */
    IOWR_32DIRECT(base, FW_RECOVERY, 1);

    /* Step 6: the caller retries anything that failed while blocked. */
    return (IORD_32DIRECT(base, FW_STATUS) & FW_ST_BLOCKED) ? -1 : 0;
}
```

Caution

The bound on step 3 is the important part. `WR_RESP_BUSY` and `RD_RESP_BUSY` clear when the peripheral delivers what it owes, and a dead peripheral never will. An unbounded poll hangs the recovery path in exactly the case it exists for. Check `WR_CMD_STUCK` and `RD_CMD_STUCK` too: those can only be cleared by `UNBLOCK`, so seeing either tells you immediately that polling `RESP_BUSY` will not be enough.

7.5 Migrating from Version 1.x

Table 29. Version 1.x to 2.0 Changes

Area	Version 1.x	Version 2.0
Peripheral reset	`m_axi_resetn` output driven by the core	Port removed; resetting the peripheral is a software step
`RESET_HOLD_CYCLES` parameter	Present	Removed
Resuming after a timeout	W1C on `STATUS.TIMEOUT_ERROR`	W1C **then** `RECOVERY.UNBLOCK`
`RECOVERY` register	Absent	New at 0x1C
`STATUS` bits 8:4	Absent	`BLOCKED`, `WR/RD_RESP_BUSY`, `WR/RD_CMD_STUCK`
`CORE_INFO` version	0x0102	0x0200

Migration steps:

1. Remove the `m_axi_resetn` connection from the Platform Designer system and provide a software-controllable reset to the peripheral instead.
2. Replace every "clear `TIMEOUT_ERROR` to resume" site with the six-step sequence in Section 7.4.
3. Gate the new path on `CORE_INFO[31:16]` if one driver must support both.

A 1.x driver run unchanged against a 2.0 core does not fail loudly. It acknowledges the fault, never writes `UNBLOCK`, and every subsequent transaction returns `SLVERR` from a core that stays blocked forever.

8. Verification

The core is verified by simulation and static elaboration. This section states precisely what has been established and what has not.

8.1 Verification Environment

Three front ends are used, because each catches things the others miss.

Table 30. Verification Tools

Tool	Version	Role	What it uniquely caught
slang	11	Strict IEEE 1800 elaboration gate	Mixed `timescale` between files
Verilator	5.48	Regression and lint, 9 parameter configurations	Lint findings across the parameter sweep
Questa	2024.1	Functional and assertion coverage	Declaration-order errors Verilator accepts

The Questa finding is worth stating plainly: three constructs in the version 2.0 RTL elaborated cleanly under Verilator and were rejected by Questa. A port connection to an identifier declared later in the file creates an implicit net that then collides with the explicit declaration. Verilator tolerated it; Questa was right to refuse. That is why the toolchain has three front ends rather than one.

8.2 Results

All figures below are from the version 2.0 sources.

Table 31. Verification Results

Metric	Result
Self-checking testbench checks	103 / 103 pass
SystemVerilog assertions	14 / 14, zero failures
Assertions with a non-vacuous pass	14 / 14
Cover directives	6 / 6 hit
FSM states covered	8 / 8
FSM transitions covered	14 / 14
`m_axi` protocol violations observed	0
Total functional coverage (Questa)	85.96 %
slang errors	0
Verilator lint findings, 9 configurations	0

Table 32. Assertions

Assertion	Passes	Checks
`a_suppress_illegal_write`	4	A denied write never reaches `m_axi`
`a_suppress_illegal_read`	4	A denied read never reaches `m_axi`
`a_err_on_blocked_write`	4	A blocked write is answered with an error, not silence
`a_err_on_blocked_read`	4	Ditto for reads
`a_awvalid_stability`	23	Slave-side `AWVALID` /payload stable until `AWREADY`
`a_arvalid_stability`	23	Slave-side `ARVALID` stability
`a_bvalid_stability`	6	`BVALID` / `BRESP` stable until `BREADY`
`a_rvalid_stability`	6	`RVALID` / `RDATA` / `RRESP` stable until `RREADY`
`a_m_awvalid_stability`	162	Master-side `AWVALID` stability, excluding the `unblock` discard
`a_m_wvalid_stability`	162	Master-side `WVALID` stability
`a_m_arvalid_stability`	119	Master-side `ARVALID` stability
`a_no_issue_while_blocked`	382	No write is issued to `m_axi` while blocked
`a_no_read_issue_while_blocked`	429	No read is issued while blocked
`a_block_holds_until_unblock`	560	`BLOCKED` persists until `UNBLOCK` , never self-clears

Every assertion has a non-zero non-vacuous pass count. This matters: an assertion that only ever passes vacuously is a comment with syntax highlighting. Two assertions in an earlier revision were exactly that, and the fix was adding a test that made the antecedent true, not editing the assertion.

Table 33. Cover Directives

Directive	Hits	What it proves the suite reaches
`c_write_denied`	4	A permission-denied write actually occurs
`c_read_denied`	4	A permission-denied read actually occurs
`c_write_decerr`	3	An out-of-range write reaching `DECERR`
`c_read_decerr`	2	An out-of-range read reaching `DECERR`
`c_block_and_recover`	5	A full block-and-recover cycle
`c_unblock_with_stuck_cmd`	3	`UNBLOCK` while a command is still stuck on `m_axi`

Table 34. FSM Transition Coverage

Transition	Write FSM	Read FSM	Exercised by
IDLE → EVAL	23	23	Every transaction reaching the core
EVAL → FWD	17	16	Every permitted transaction
EVAL → RESP	5	6	Denied, out-of-range and blocked transactions
EVAL → IDLE	1	1	Reset asserted during EVAL
FWD → RESP	14	13	Normal completion and timeout
FWD → IDLE	3	3	Reset asserted during FWD
RESP → IDLE	19	19	Every completed response

8.3 The Orphan-Response Measurement

`verification/orphan_response_tb.sv` measures the hazard that motivates step 4 of the recovery sequence. It runs a clean, permitted write, injects one late `BVALID` from an abandoned transaction at offset *k* cycles, and sweeps *k* over 25 values. If the master ever sees `SLVERR` on a write the peripheral answered `OKAY`, a stale response was mis-attributed.

Table 35. Orphan-Response Results

Recovery sequence	Offsets that mis-attribute
Peripheral reset before `UNBLOCK` (documented)	0 of 25
`UNBLOCK` without resetting the peripheral	1 of 25

This is why step 4 is stated as mandatory rather than recommended.

8.4 What Has Not Been Verified

Stated plainly, because the absence of a result is easy to mistake for a passing one.

Table 36. Not Verified

Item	Status
Quartus Prime analysis of the `_hw.tcl` component	Never run. The component has not been imported into Platform Designer.
Synthesis results (logic elements, registers)	Never run
Timing closure, f_{MAX} vs `NUM_RULES`	Never run
Behaviour inside a generated Platform Designer interconnect	Never run
Any specific device family	None characterised
Formal property proof	Not attempted; assertions are simulation-only

Everything in Sections 1 through 7 that describes core behaviour is backed by simulation. Everything about *building* the core into a real system is untested and should be treated as a first attempt.

9. Design Considerations and Limitations

9.1 Single Outstanding Transaction

Each datapath handles one transaction at a time. A read and a write may overlap, but a second write waits for the first to complete. Interconnect generated by Platform Designer will pipeline requests into the firewall, which simply back-pressures.

This is a deliberate simplification. The core answers the master itself on a timeout, and doing that correctly with multiple outstanding transactions requires tracking which responses are still owed and which have been synthesised — the exact bookkeeping that `RESP_BUSY` and `CMD_STUCK` expose in rudimentary form today.

9.2 No Autonomous Flushing

When blocked, the core stops forwarding but does not itself complete or discard transactions the peripheral already accepted. The consequence is the bounded poll in Section 3.5.1.

AMD's AXI Protocol Firewall does flush autonomously, which is why its busy bits always reach zero and its recovery has no equivalent caveat. Adding this is the most valuable planned enhancement.

9.3 Address-Only Policy

Rules key on address range and direction. `AWPROT` / `ARPROT` are forwarded but not evaluated, so the core cannot express "privileged accesses only" or "instruction fetches only".

9.4 No Protocol Legality Checking

The core does not police AXI legality on `s_axi`. A master that violates VALID stability will not be caught here. This is the main functional difference from AMD's core, which is a protocol checker with no address-based access control. The two are complementary.

Table 37. Comparison with the AMD AXI Protocol Firewall

Capability	This core	AMD AXI Protocol Firewall
Address-range access control	Yes, up to 64 rules	No
Read/write permission per range	Yes	No
AXI protocol legality checking	No	Yes, extensive
Timeout on the protected side	Yes	Yes
Guaranteed response to the master	Yes	Yes
Blocks after a fault until unblocked	Yes	Yes
Autonomous flushing when blocked	No	Yes
Requires resetting the monitored side before unblock	Yes	Yes
Full AXI4 (bursts, IDs)	No, AXI4-Lite only	Yes

9.5 Rule Table Is Not Lockable

Any master that can reach `s_axi_ctrl` can rewrite the rule table, including disabling the firewall through `GLOBAL_ENABLE`. The core protects a peripheral from a misbehaving master on `s_axi`; it does not protect itself from a malicious master on `s_axi_ctrl`. If your threat model includes the latter, place the control port behind something that arbitrates access to it — a second firewall instance is one option.

A one-way lock bit is a natural addition and is not currently implemented.

9.6 Combinational Rule Lookup

The lookup is a combinational priority chain over `NUM_RULES` entries, so it is the expected critical path and it scales with `NUM_RULES`. With no synthesis data there is no guidance on where it stops meeting timing. If it limits f_{MAX}, register the lookup result and accept one more cycle in EVAL.

9.7 Shared Fault Capture

`FAULT_ADDR` and `FAULT_INFO` are single registers shared by both datapaths. Simultaneous read and write faults both set their sticky `STATUS` bits correctly, but only one address and one type are captured — and the two are resolved by *different* rules:

Table 38. Fault Capture Resolution

Field	Resolved by
`FAULT_ADDR`	Write side wins
`FAULT_INFO.WAS_WRITE`	Write side wins
`FAULT_INFO.FAULT_TYPE`	Type precedence across both datapaths: `TIMEOUT` > `PERM` > `ADDR`

The two rules agree whenever both datapaths fault with the same type, or when only one faults. They disagree when the datapaths fault in the same cycle with different types: the address and direction then describe the write while the type may describe the read.

This is a consequence of the capture being one register pair fed by three independent fault sources, and it is not currently arbitrated. Software that must attribute every fault individually needs to correlate with its own transaction log. Giving each datapath its own capture pair would remove the ambiguity and is a candidate for a future revision.

9.8 Synchronous Reset Only

`resetn` is sampled synchronously. If your reset source is asynchronous, put a reset synchroniser ahead of the core. Platform Designer's reset bridge does this for you.

10. Document Revision History

Table 39. Document Revision History

Document version	Core version	Date	Changes
1.0	2.0	August 2026	Initial release of this user guide

Table 40. Core Revision History

Core version	`CORE_INFO`	Changes
2.0	0x0200	Removed the `m_axi_resetn` output and the `RESET_HOLD_CYCLES` parameter. Added the `RECOVERY` register with `UNBLOCK`, and `STATUS` bits `BLOCKED`, `WR/RD_RESP_BUSY` and `WR/RD_CMD_STUCK`. Recovery became an explicit software sequence. Sources converted to SystemVerilog. **Breaking change.**
1.2	0x0102	Fixed two AXI compliance defects on the control port: `AWREADY` / `WREADY` were asserted while a `BVALID` was outstanding, and `ARREADY` while an `RVALID` was, so a pipelining master lost responses. Fixed a rule-register byte-merge defect when `ADDR_WIDTH` < 32. Removed dead discard-pending logic.
1.1	0x0101	Split the single access-violation assertion into separate read and write assertions, which exposed that no test had ever denied a read.
1.0	0x0100	Initial release